EXPERIMENT 10

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Experiment 10 – Design of Trigger for Automatic Primary Key Generation (Enterprise Application)

Experiment

Designing and implementing a database trigger that automatically generates and enforces unique primary key values during record insertion. This experiment demonstrates automation, data integrity, and elimination of manual errors in enterprise database systems.

Aim

To create and implement a database trigger that automatically generates a unique primary key for each record inserted into a table.

Objective

- To understand the concept and working of database triggers
- To implement automatic primary key generation using triggers
- To eliminate manual intervention in key assignment
- To ensure uniqueness and data integrity
- To apply trigger-based automation in enterprise systems

Software Requirements

Database Management System:

- Oracle Database Express Edition (Oracle XE) / PostgreSQL

Database Administration Tool:

- Oracle SQL Developer / pgAdmin

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Problem Statement

In enterprise-level applications, manually assigning primary keys can lead to duplication and errors. To maintain data integrity and ensure uniqueness, a trigger-based mechanism is required that automatically generates a primary key before inserting records into a table.

Practical / Experiment Steps

- Create a table requiring automatic primary key generation
- Create a sequence to generate unique values
- Design a trigger that activates before insert
- Assign the next sequence value to the primary key
- Insert multiple records to test the trigger
- Verify uniqueness of generated keys

Procedure

- Open Oracle SQL Developer and connect to the database
- Create the required table
- Create a sequence for primary key generation
- Create a trigger using BEFORE INSERT event
- Compile the trigger successfully
- Insert records without specifying the primary key
- Display the table contents

Input / Output Details

Input

Table: employee (emp_id, name, salary)

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Step-1: Create EMPLOYEE Table

[SQL Worksheet]*

```
CREATE TABLE employee (
emp_id INT PRIMARY KEY,
name VARCHAR(50),
salary INT
);

CREATE SEQUENCE emp_seq
START WITH 1
INCREMENT BY 1;

CREATE OR REPLACE TRIGGER emp_trigger
BEFORE INSERT ON employee
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> CREATE TABLE employee (
     emp_id INT PRIMARY KEY,
     name VARCHAR(50),
     salary INT...
Show more...


Table EMPLOYEE created.

Elapsed: 00:00:00.021
```

Step-2: Create Sequence emp_seq

[SQL Worksheet]*

```
name VARCHAR(50),
salary INT
);

CREATE SEQUENCE emp_seq
START WITH 1
INCREMENT BY 1;

CREATE OR REPLACE TRIGGER emp_trigger
BEFORE INSERT ON employee
FOR EACH ROW
BEGIN
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> CREATE SEQUENCE emp_seq
     START WITH 1
     INCREMENT BY 1


Sequence EMP_SEQ created.

Elapsed: 00:00:00.007
```

Step-3: Create Trigger emp_trigger

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[SQL Worksheet]*

```
CREATE OR REPLACE TRIGGER emp_trigger
BEFORE INSERT ON employee
FOR EACH ROW
BEGIN
IF :NEW.emp_id IS NULL THEN
SELECT emp_seq.NEXTVAL INTO :NEW.emp_id FROM dual;
END IF;
END;
/

INSERT INTO employee (name, salary) VALUES ('Aman', 40000);
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> CREATE OR REPLACE TRIGGER emp_trigger
     BEFORE INSERT ON employee
     FOR EACH ROW
     BEGIN...
Show more...


Trigger EMP_TRIGGER compiled

Elapsed: 00:00:00.022
```

Step-4: Insert data into table

[SQL Worksheet]*

```
END;
/

INSERT INTO employee (name, salary) VALUES ('Aman', 40000);
INSERT INTO employee (name, salary) VALUES ('Neha', 50000);
INSERT INTO employee (name, salary) VALUES ('Rahul', 45000);
INSERT INTO employee (name, salary) VALUES ('Simran', 55000);

SELECT * FROM employee;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
Elapsed: 00:00:00.002

SQL> INSERT INTO employee (name, salary) VALUES ('Simran', 55000)


1 row inserted.

Elapsed: 00:00:00.001
```

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Step-5: Show table data

[SQL Worksheet]*

```
END;
/

INSERT INTO employee (name, salary) VALUES ('Aman', 40000);
INSERT INTO employee (name, salary) VALUES ('Neha', 50000);
INSERT INTO employee (name, salary) VALUES ('Rahul', 45000);
INSERT INTO employee (name, salary) VALUES ('Simran', 55000);

SELECT * FROM employee;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.131 seconds

	EMP_ID	NAME	SALARY
1	1	Aman	40000
2	2	Neha	50000
3	3	Rahul	45000
4	4	Simran	55000

SQL Implementation Overview

- Table creation
- Sequence creation
- Trigger creation
- Data insertion
- Result verification

Output

After executing the program, each record is automatically assigned a unique primary key without manual input.

Example Output:
1 Aman 40000
2 Neha 50000
3 Rahul 45000
4 Simran 55000

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Learning Outcome

After completing this experiment, the learner will be able to:

- Understand the working of database triggers
- Implement automatic primary key generation
- Ensure data integrity without manual intervention
- Apply trigger-based solutions in enterprise applications
- Develop efficient and reliable database systems